

1.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

```
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc Week1_1.c -o Week1_1
rehan_lenovo@Rehan-Lenovo:~/OS$ ./Week1_1
Enter Linux command: uname
Parent PID: 697
Child PID: 698
Linux
Child process completed.
rehan_lenovo@Rehan-Lenovo:~/OS$ |
```

2. copying the contents from one file to another file

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
rehan_lenovo@Rehan-Lenovo:~/OS$ ls
Exec.c Fork.c PID.c Parent.c Wait.c
rehan_lenovo@Rehan-Lenovo:~/OS$ nano Week1_2.c
rehan_lenovo@Rehan-Lenovo:~/OS$ nano input.txt
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc Week1_2.c -o Week1_2
rehan_lenovo@Rehan-Lenovo:~/OS$ ./Week1_2
File copied successfully.
rehan_lenovo@Rehan-Lenovo:~/OS$ nano output.txt
rehan_lenovo@Rehan-Lenovo:~/OS$ |
```